

LAB 1: SQL ON ALCOHOL SALES

QUESTIONS

1. How many total products are in the Products table?

Answer: There are 9,977 products in the Products table.

In this dataset, a product is defined by a unique `item_no`, which represents a distinct catalog listing rather than just a physical item. Multiple records may share the same `item_description`, `category_name`, `bottle_size`, and packaging attributes (`pack` and `inner_pack`) but still be treated as separate products due to differences in vendor (`vendor_name`), prices (`bottle_price`, `shelf_price`, and `case_cost`), or regulatory listing (`list_date`). Therefore, even visually identical items can correspond to different product entries in the system.

--Count number of unique products (by `item_no`)

```
SELECT COUNT(DISTINCT item_no) AS unique_products
FROM products
```

	unique_products
	bigint
1	9977
Total rows: 1 Query complete 00:00:00.666	

/*Additional verification queries*/

--1. Check for products of the same `category_name` & `item_description` with multiple `list_dates`

```
SELECT
    category_name,
    item_description,
    bottle_size,
    pack,
    MIN(list_date) AS earliest_list_date,
    MAX(list_date) AS latest_list_date,
    COUNT(DISTINCT item_no) AS number_of_listings
FROM products
GROUP BY
    category_name,
    item_description,
    bottle_size,
    pack
HAVING COUNT(DISTINCT item_no) > 1
ORDER BY number_of_listings DESC;
--600 records
```

	category_name text	item_description text	bottle_size integer	pack integer	earliest_list_date timestamp without time zone	latest_list_date timestamp without time zone	number_of_listings bigint
1	DECANTERS & SPECIALTY PACKAGES	Knob Creek Gift Set	750	6	1997-10-01 00:00:00	2003-08-01 00:00:00	4
2	TEQUILA	Gran Centenario Anejo	750	6	2005-09-01 00:00:00	2007-10-10 00:00:00	3
3	FLAVORED VODKA	Smirnoff Coconut	750	12	2013-09-13 00:00:00	2014-07-01 00:00:00	3
4	TRIPLE SEC	Arrow Triple Sec	1000	12	1992-10-13 00:00:00	2014-05-15 00:00:00	3
5	AMERICAN DRY GINS	Seagram's Distiller's Reserve Gin	750	12	2006-05-01 00:00:00	2013-12-05 00:00:00	3
6	DECANTERS & SPECIALTY PACKAGES	Canadian Club Glass Set	750	6	1989-10-01 00:00:00	1998-10-01 00:00:00	3
7	MISCELLANEOUS SCHNAPPS	99 Oranges	750	12	2004-08-01 00:00:00	2010-02-24 00:00:00	3
8	FLAVORED VODKA	Skinnygirl Tangerine	750	12	2012-04-01 00:00:00	2014-11-01 00:00:00	3
9	CANADIAN WHISKIES	Pendleton Canadian Whisky	1750	6	2009-01-19 00:00:00	2011-01-01 00:00:00	3
10	DECANTERS & SPECIALTY PACKAGES	Chambord Liqueur Glass Set	375	6	1996-09-01 00:00:00	1998-09-01 00:00:00	3
11	FLAVORED VODKA	Smirnoff Twist Of Watermelon	502	12	2004-06-01 00:00:00	2009-12-04 00:00:00	3
12	DECANTERS & SPECIALTY PACKAGES	After Shock Gift Set	750	6	1995-10-01 00:00:00	2001-08-01 00:00:00	3
13	IMPORTED DRY GINS	Bombay Sapphire Gin	200	12	2004-06-01 00:00:00	2014-07-03 00:00:00	3
14	IMPORTED DRY GINS	Bombay Sapphire Gin	375	12	1997-11-01 00:00:00	2004-06-01 00:00:00	3
15	SCOTCH WHISKIES	Muirhead Scotch	1000	12	1995-11-01 00:00:00	2000-08-01 00:00:00	3
16	CANADIAN WHISKIES	Revelstoke Spiced Canadian	1000	12	2000-09-01 00:00:00	2006-06-01 00:00:00	3
17	CREAM LIQUEURS	Prichard's Fudge Brownie Cream Liqueur	750	6	2012-12-04 00:00:00	2013-03-01 00:00:00	3
18	TEQUILA	Patron Reposado	375	12	2007-04-12 00:00:00	2014-02-01 00:00:00	3
19	IMPORTED VODKA	Blavod Black Vodka	750	6	2005-05-01 00:00:00	2009-01-02 00:00:00	3
20	TEQUILA	El Jimador Reposado Tequila	750	12	2000-01-01 00:00:00	2001-01-01 00:00:00	3
Total rows: 600		Query complete 00:00:00.604					

--Check for sample record of the same category_name & item_description with multiple values of list_date

```
SELECT *
FROM products
WHERE category_name = 'IMPORTED DRY GINS'
AND item_description = 'Bombay Sapphire Gin'
ORDER BY bottle_size, pack, list_date DESC
--12 records
```

Item_no [PK] integer	category_name text	Item_description text	vendor integer	vendor_name text	bottle_size integer	pack integer	inner_pack integer	age text	proof text	list_date timestamp without time zone	upc text	acc text	bottle_price money	shelf_price numeric (10,2)	case_cost numeric (10,2)
1	IMPORTED DRY GINS	Bombay Sapphire Gin	35	Bacardi U.s.a. Inc.	50	120	1	[null]	94	1997-02-01 00:00:00	[null]	[null]	\$0.84	1.26	100.48
2	IMPORTED DRY GINS	Bombay Sapphire Gin	35	Bacardi U.s.a. Inc.	200	12	1	[null]	94	2014-07-03 00:00:00	80480000066	928223	\$4.00	6.00	47.90
3	IMPORTED DRY GINS	Bombay Sapphire Gin	35	Bacardi U.s.a. Inc.	200	12	1	[null]	94	2004-06-01 00:00:00	[null]	[null]	\$3.49	5.24	41.90
4	IMPORTED DRY GINS	Bombay Sapphire Gin	35	Bacardi U.s.a. Inc.	200	12	1	[null]	94	2004-06-01 00:00:00	80480301149	80480046095	\$3.99	5.99	47.90
5	IMPORTED DRY GINS	Bombay Sapphire Gin	35	Bacardi U.s.a. Inc.	375	12	1	[null]	94	2004-06-01 00:00:00	[null]	80480046088	\$3.57	5.36	42.80
6	IMPORTED DRY GINS	Bombay Sapphire Gin	35	Bacardi U.s.a. Inc.	375	12	1	[null]	94	2004-06-01 00:00:00	[null]	[null]	\$6.66	9.99	79.94
7	IMPORTED DRY GINS	Bombay Sapphire Gin	35	Bacardi U.s.a. Inc.	375	12	1	[null]	80	1997-11-01 00:00:00	[null]	902979	\$5.41	8.12	64.95
8	IMPORTED DRY GINS	Bombay Sapphire Gin	35	Bacardi U.s.a. Inc.	375	24	1	[null]	94	1998-09-01 00:00:00	[null]	[null]	\$6.33	9.50	151.84
9	IMPORTED DRY GINS	Bombay Sapphire Gin	35	Bacardi U.s.a. Inc.	750	12	1	[null]	94	1990-04-01 00:00:00	80480301026	1008050000000	\$13.64	20.46	163.57
10	IMPORTED DRY GINS	Bombay Sapphire Gin	35	Bacardi U.s.a. Inc.	1000	12	1	[null]	94	1988-05-01 00:00:00	80480301019	1008050000000	\$15.61	24.42	187.28
11	IMPORTED DRY GINS	Bombay Sapphire Gin	35	Bacardi U.s.a. Inc.	1750	6	1	[null]	94	2000-02-01 00:00:00	80480301002	1008050000000	\$26.22	39.33	157.30
12	IMPORTED DRY GINS	Bombay Sapphire Gin	35	Bacardi U.s.a. Inc.	1750	6	1	[null]	86	1996-11-10 00:00:00	[null]	[null]	\$21.97	32.96	131.79
Total rows: 12		Query complete 00:00:00.644						Rows selected: 3		LF Ln 203, Col 61					

Note: Highlighted in blue is a scenario where each record has a different item_no where additional fields have the same value bottle_size=375 and pack=12 and inner_pack=1 with multiple values of proof, list_dates, and prices (bottle_price, shelf_price and case_cost)

--2. Check for products of the same category_name & item_description with multiple multiple values of inner_pack

```
SELECT
    category_name,
    item_description,
    bottle_size,
    pack,
    COUNT(DISTINCT inner_pack) AS number_of_inner_pack
FROM products
```

```

GROUP BY
    category_name,
    item_description,
    bottle_size,
    pack
HAVING COUNT(DISTINCT inner_pack) > 1
ORDER BY number_of_inner_pack DESC;
--3 records

```

	category_name text	item_description text	bottle_size integer	pack integer	number_of_inner_pack bigint
1	80 PROOF VODKA	River Pilot Vodka	750	6	2
2	FLAVORED VODKA	Smirnoff Twist Of Watermelon	502	12	2
3	SCOTCH WHISKIES	J & B Rare Scotch	50	10	2
Total rows: 3 Query complete 00:00:00.695					

```

--Check for sample record with multiple inner_packs
SELECT *
FROM products
WHERE category_name = 'FLAVORED VODKA'
AND item_description = 'Smirnoff Twist Of Watermelon'
ORDER BY bottle_size, pack, list_date DESC
--6 records

```

	item_no [PK] integer	category_name text	item_description text	vendor integer	vendor_name text	bottle_size integer	pack integer	inner_pack integer	age text	proof text	list_date timestamp without time zone	upc text	scc text	bottle_price money	shelf_price numeric (10,2)	case_cost numeric (10,2)
1	904820	FLAVORED VODKA	Smirnoff Twist Of Watermelon	260	Diageo Americas	375	12	1	[null]	70	2010-04-19 00:00:00	82000001263	82000767985	\$4.50	6.75	53.95
2	904610	FLAVORED VODKA	Smirnoff Twist Of Watermelon	260	Diageo Americas	502	12	10	[null]	70	2009-12-04 00:00:00	82000001164	82000767961	\$5.72	8.58	68.58
3	40941	FLAVORED VODKA	Smirnoff Twist Of Watermelon	260	Diageo Americas	502	12	1	[null]	70	2004-06-01 00:00:00	[null]	[null]	\$5.72	8.58	68.58
4	40944	FLAVORED VODKA	Smirnoff Twist Of Watermelon	260	Diageo Americas	502	12	1	[null]	70	2004-06-01 00:00:00	[null]	[null]	\$3.38	5.07	40.59
5	904947	FLAVORED VODKA	Smirnoff Twist Of Watermelon	260	Diageo Americas	1000	12	1	[null]	70	2009-01-13 00:00:00	82000001188	82000768005	\$8.96	13.44	107.42
6	40947	FLAVORED VODKA	Smirnoff Twist Of Watermelon	260	Diageo Americas	1000	12	1	[null]	70	2009-01-13 00:00:00	[null]	[null]	\$9.20	13.80	110.43
Total rows: 6 Query complete 00:00:00.648 Rows selected: 3																
Ln 236, Col 1																

Note: Highlighted in blue is a scenario where each record has a different item_no where additional fields have the same value bottle_size=502 and pack=12 with multiple values of inner_pack, proof, list_dates, and prices (bottle_price, shelf_price, and case_cost)

2. Who are the top most diverse vendors (i.e. they have the highest number of distinct products)? How many different products do they have?

Answer: The most diverse vendors, based on the number of distinct products, are:

1. Jim Beam Brands – 925 products
2. Diageo Americas – 907 products
3. Pernod Ricard USA / Austin Nichols – 599 products
4. Yahara Bay Distillers Inc – 579 products
5. Heaven Hill Distilleries Inc. – 388 products
6. Bacardi U.S.A. Inc. – 357 products
7. Luxco (St. Louis) – 333 products
8. MHW Ltd – 310 products
9. Sazerac Co. Inc. – 281 products
10. Sazerac North America – 265 products

```

SELECT vendor,
       vendor_name,
       COUNT(DISTINCT item_no) AS number_of_products
FROM products
GROUP BY vendor, vendor_name
ORDER BY number_of_products DESC
LIMIT 10;

```

	vendor integer	vendor_name text	number_of_products bigint
1	65	Jim Beam Brands	925
2	260	Diageo Americas	907
3	370	Pernod Ricard Usa/austin Nichols	599
4	885	Yahara Bay Distillers Inc	579
5	259	Heaven Hill Distilleries Inc.	388
6	35	Bacardi U.s.a. Inc.	357
7	434	Luxco-st Louis	333
8	305	Mhw Ltd	310
9	421	Sazerac Co. Inc.	281
10	55	Sazerac North America	265
Total rows: 10		Query complete 00:00:00.681	

The results show that **a small number of large vendors dominate product diversity**, with **Jim Beam Brands and Diageo Americas each offering over 900 distinct products**—significantly higher than the rest. There is a noticeable drop after the top two vendors, suggesting a concentration of product variety among a few major suppliers.

The presence of both global spirits companies (e.g., Diageo, Pernod Ricard, Bacardi) and regional or smaller distillers (e.g., Yahara Bay Distillers) indicates that **the market includes a mix of large-scale distributors and niche producers**, although the largest vendors still account for the greatest product breadth.

*/*Additional verification queries*/*

--Checking the uniqueness of the vendor IDs and vendor names

```

SELECT COUNT(DISTINCT vendor) AS number_of_vendorIDs
FROM products

```

--271 vendor IDs

number_of_vendorids bigint
271

```

SELECT COUNT(DISTINCT (vendor, vendor_name)) AS number_of_vendors
FROM products

```

--271 vendors

number_of_vendors bigint
271

```

SELECT COUNT(DISTINCT vendor_name) AS number_of_vendornames
FROM products

```

--271 vendor names

number_of_vendornames
bigint
271

--Check sample vendor and products

```
SELECT *
FROM products
WHERE vendor_name = 'Mhw Ltd'
ORDER BY category_name, item_description, bottle_size, pack
```

	item_no [PK] integer	category_name text	item_description text	vendor integer	vendor_name text	bottle_size integer	pack integer	inner_pack integer	age text	proof text	lis tin
1	36006	80 PROOF VODKA	Fuzzy's Ultra Premium Vodka	305	Mhw Ltd	750	6	1	[null]	80	2C
2	36065	80 PROOF VODKA	Fuzzy's Ultra Premium Vodka	305	Mhw Ltd	1750	3	1	[null]	80	2C
3	904720	80 PROOF VODKA	Fuzzy Vodka	305	Mhw Ltd	750	6	1	[null]	80	2C
4	902083	80 PROOF VODKA	Zodiac Vodka(dno)	305	Mhw Ltd	750	12	1	[null]	80	2C
5	72596	AMERICAN AMARETTO	Bols Amaretto	305	Mhw Ltd	750	12	1	[null]	56	2C
6	33337	AMERICAN SLOE GINS	Bols Sloe Gin	305	Mhw Ltd	1000	12	1	[null]	60	1S
7	82510	APPLE SCHNAPPS	Bols Sour Apple Schnapps	305	Mhw Ltd	1000	12	1	[null]	34	1S
8	54096	APRICOT BRANDIES	Bols Apricot Brandy	305	Mhw Ltd	750	12	1	[null]	70	1S
9	902827	APRICOT BRANDIES	Bols Apricot Brandy	305	Mhw Ltd	750	12	1	[null]	70	2C
10	54097	APRICOT BRANDIES	Bols Apricot Brandy	305	Mhw Ltd	1000	12	1	[null]	70	1S
11	902613	BARBADOS RUM	Barbancourt 3* Rhum 4yr	305	Mhw Ltd	750	12	1	[null]	86	1S
12	902594	BARBADOS RUM	Barbancourt 5* Rhum 8yr	305	Mhw Ltd	750	12	1	[null]	86	1S
Total rows: 310 Query complete 00:00:00.642											LF Ln 255, Col 14

3. Which products sell the best by total number of unit sales?

Answer: The products that sell the best by total number of units (bottles sold) are:

1. Black Velvet (Canadian Whiskies) – 828,393 bottles
2. Hawkeye Vodka (80 Proof Vodka) – 500,070 bottles
3. Captain Morgan Spiced Rum (Spiced Rum) – 483,621 bottles
4. McCormick Vodka PET (80 Proof Vodka) – 393,203 bottles
5. Barton Vodka (80 Proof Vodka) – 288,425 bottles
6. Fireball Cinnamon Whiskey (Whiskey Liqueur) – 279,771 bottles
7. Captain Morgan Original Spiced (Spiced Rum) – 263,490 bottles
8. McCormick Vodka (80 Proof Vodka) – 259,268 bottles
9. Jack Daniels Old No. 7 Black Label (Tennessee Whiskey) – 250,408 bottles
10. Fireball Cinnamon Whiskey (Whiskey Liqueur) – 246,633 bottles

```
SELECT item,
       category_name,
       description,
       SUM(bottle_qty) AS sum_bottle_qty,
       SUM(total) AS sum_sales
FROM sales
GROUP BY item, category_name, description
ORDER BY sum_bottle_qty DESC
LIMIT 10;
```

	item integer	category_name text	description text	sum_bottle_qty bigint	sum_sales numeric
1	11788	CANADIAN WHISKIES	Black Velvet	828393	12863376.81
2	36308	80 PROOF VODKA	Hawkeye Vodka	500070	5380753.20
3	43337	SPICED RUM	Captain Morgan Spiced Rum	483621	8523370.55
4	36904	80 PROOF VODKA	Mccormick Vodka Pet	393203	1059371.74
5	35317	80 PROOF VODKA	Barton Vodka	288425	1717312.26
6	64867	WHISKEY LIQUEUR	Fireball Cinnamon Whiskey	279771	4758904.71
7	43336	SPICED RUM	Captain Morgan Original Spiced	263490	3497803.08
8	36903	80 PROOF VODKA	Mccormick Vodka	259268	440755.60
9	26827	TENNESSEE WHISKIES	Jack Daniels Old #7 Black Lbl	250408	6668863.94
10	64866	WHISKEY LIQUEUR	Fireball Cinnamon Whiskey	246633	3322146.51
Total rows: 10		Query complete 00:00:01.417			

The results show that **vodka and whiskey-based products dominate unit sales**, with multiple entries from 80-proof vodka brands and whiskey variants appearing in the top rankings. Black Velvet leads by a significant margin, indicating strong demand for certain high-volume, likely value-oriented products.

There is a strong concentration of **80-proof vodka products** in the Top 10, suggesting that standard-strength vodka is a high-volume category in the market.

Some products appear multiple times due to different item listings (e.g., packaging variations or product versions), reinforcing that unit sales are aggregated at the `item_no` level rather than the brand level:

- Fireball Cinnamon Whiskey appearing twice (Top 6 and 10)
- McCormick Vodka appearing twice (Top 4 and 8)

*/*Additional verification queries*/*

--Check if there are multiple product descriptions for 1 item

```
SELECT item, COUNT(DISTINCT description) AS multiple_description
FROM sales
GROUP BY item
HAVING COUNT(DISTINCT description) > 1
--135 items
```

	item integer	multiple_description bigint
1	874	2
2	2666	2
3	2918	2
4	4032	2
5	4131	2
6	4556	2
7	4866	2
8	4867	2
9	4868	2
10	4876	2
Total rows: 135		Query complete 00:00:03.536

--Check for sample record

```
SELECT DISTINCT date, item, description
FROM sales
WHERE item = 874
ORDER BY date
```

--60 sales records

	date timestamp without time zone	item integer	description text
1	2014-02-12 00:00:00	874	Glenlivet w/2-50mls
2	2014-02-20 00:00:00	874	Glenlivet w/2-50mls
3	2014-02-24 00:00:00	874	Glenlivet w/2-50mls
4	2014-03-03 00:00:00	874	Glenlivet w/2-50mls
5	2014-03-05 00:00:00	874	Glenlivet w/2-50mls
6	2014-03-10 00:00:00	874	Glenlivet w/2-50mls
7	2014-05-14 00:00:00	874	The Glenlivet w/50ml 15YR & 18YR
8	2014-05-20 00:00:00	874	The Glenlivet w/50ml 15YR & 18YR
9	2014-05-22 00:00:00	874	The Glenlivet w/50ml 15YR & 18YR
10	2014-05-23 00:00:00	874	The Glenlivet w/50ml 15YR & 18YR
11	2014-05-28 00:00:00	874	The Glenlivet w/50ml 15YR & 18YR
12	2014-05-29 00:00:00	874	The Glenlivet w/50ml 15YR & 18YR
Total rows: 60		Query complete 00:00:00.700	

--Products appearing twice

```
SELECT *
FROM products
WHERE item_no IN (36903, 36904, 64866, 64867)
ORDER BY item_no
```

item_no [PK] integer	category_name text	item_description text	vendor integer	vendor_name text	bottle_size integer	pack integer	inner_pack integer	age text	proof text	list_date timestamp without time zone	upc text	scc text	bottle_price money	shelf_price numeric (10,2)	case_cost numeric (10,2)
1	36903	80 PROOF VODKA	McCormick Vodka	300	McCormick Distilling Company	200	48	1	[null]	80	2004-03-01 00:00:00	85592104572	10085600000000	\$1.13	1.70
2	36904	80 PROOF VODKA	McCormick Vodka Pet	300	McCormick Distilling Company	375	24	1	[null]	80	1992-11-01 00:00:00	85592104565	10085600000000	\$1.80	2.70
3	64866	WHISKEY LIQUEUR	Fireball Cinnamon Whiskey	421	Sazerac Co. Inc.	750	12	1	[null]	80	2001-04-01 00:00:00	88004146689	10088000000000	\$8.98	13.47
4	64867	WHISKEY LIQUEUR	Fireball Cinnamon Whiskey	421	Sazerac Co. Inc.	1000	12	1	[null]	80	2001-04-01 00:00:00	88004144678	10088000000000	\$11.34	17.01
Total rows: 4		Query complete 00:00:00.672													

Note: Products with the same category_name and item_description can have different values for item_no due to differences in packaging attributes (bottle_size, pack, inner_pack), pricing (bottle_price, shelf_price and case_cost), or regulatory listing (list_date).

4. Which products sell the best by total dollar value of sales?

Answer: The Top 10 products that generate the highest total sales revenue are:

1. Black Velvet (Canadian Whiskies) – \$12,863,376.81
2. Captain Morgan Spiced Rum (Spiced Rum) – \$8,523,370.55
3. Jack Daniels Old No. 7 Black Label (Tennessee Whiskies) – \$6,668,863.94
4. Crown Royal Canadian Whisky (Canadian Whiskies) – \$6,231,299.41
5. Hawkeye Vodka (80 Proof Vodka) – \$5,380,753.20
6. Fireball Cinnamon Whiskey (Whiskey Liqueur) – \$4,758,904.71
7. Captain Morgan Spiced Rum (Spiced Rum) – \$4,341,131.70
8. Crown Royal (Canadian Whiskies) – \$4,056,735.73
9. Jack Daniels Old No. 7 Black Label (Tennessee Whiskies) – \$3,861,152.90
10. Captain Morgan Spiced Rum (Spiced Rum) – \$3,774,545.06

```

SELECT item,
       category_name,
       description,
       SUM(bottle_qty) AS sum_bottle_qty,
       SUM(total) AS total_sales
FROM sales
GROUP BY item, category_name, description
ORDER BY total_sales DESC
LIMIT 10;

```

	item integer	category_name text	description text	sum_bottle_qty bigint	total_sales numeric
1	11788	CANADIAN WHISKIES	Black Velvet	828393	12863376.81
2	43337	SPICED RUM	Captain Morgan Spiced Rum	483621	8523370.55
3	26827	TENNESSEE WHISKIES	Jack Daniels Old #7 Black Lbl	250408	6668863.94
4	11297	CANADIAN WHISKIES	Crown Royal Canadian Whis...	226493	6231299.41
5	36308	80 PROOF VODKA	Hawkeye Vodka	500070	5380753.20
6	64867	WHISKEY LIQUEUR	Fireball Cinnamon Whiskey	279771	4758904.71
7	43338	SPICED RUM	Captain Morgan Spiced Rum	162180	4341131.70
8	11296	CANADIAN WHISKIES	Crown Royal	181894	4056735.73
9	26826	TENNESSEE WHISKIES	Jack Daniels Old #7 Black Lbl	178738	3861152.90
10	1799	SPICED RUM	Captain Morgan Spiced Barr...	141005	3774545.06
Total rows: 10		Query complete 00:00:01.274			

While high-volume products still perform strongly, the revenue rankings are more **dominated by well-known premium brands**, such as Captain Morgan, Black Velvet and Jack Daniels. This indicates that **price plays a significant role in revenue generation**, not just sales volume.

Compared to unit sales, where vodka dominated, revenue rankings show a stronger presence of **rum and whiskey**, suggesting that **higher-priced products contribute more significantly to total revenue despite lower sales volume**.

Canadian whiskies and established global brands such as Jack Daniels, Captain Morgan, Crown Royal consistently appear in the top rankings, indicating strong brand equity and pricing power in driving revenue.

Multiple entries for the same product reflect different product listings or packaging variants, reinforcing that revenue is calculated at the individual item level (`item_no`) rather than aggregated at the brand level (`category_name` and `description`).

1. Captain Morgan Spiced Rum (Top 2 and 7)
2. Crown Royal (Top 4 and 8)
3. Jack Daniels Old No. 7 (Top 3 and 9)

*/*Additional verification queries*/*

```

--Check for products appearing twice for Captain Morgan, Crown Royal, Jack Daniels
SELECT *
FROM products
WHERE item_no IN (43337, 1799, 11297, 11296, 26827, 26826)
ORDER BY item_description

```

	item_no [PK] integer	category_name text	item_description text	vendor integer	vendor_name text	bottle_size integer	pack integer	inner_pack integer	age text	proof text	list_date timestamp without time zone	upc text	scc text	bottle_price money	shelf_price numeric (10,2)	cas num
1	1799	SPICED RUM	Captain Morgan Spiced Barr...	260	Diageo Americas	1750	6	1	[null]	70	2013-03-01 00:00:00	82000759102	10082000000000	\$18.00		27.00
2	43337	SPICED RUM	Captain Morgan Spiced Rum	260	Diageo Americas	1000	12	1	[null]	70	1990-10-01 00:00:00	87000002708	82000747178	\$11.75		17.63
3	11296	CANADIAN WHISKIES	Crown Royal	260	Diageo Americas	750	12	1	[null]	80	1972-04-29 00:00:00	87000007253	82000736936	\$15.00		22.50
4	11297	CANADIAN WHISKIES	Crown Royal Canadian Whis...	260	Diageo Americas	1000	12	1	[null]	80	1988-08-01 00:00:00	87000007246	10087000000000	\$18.50		27.75
5	26826	TENNESSEE WHISKIES	Jack Daniels Old #7 Black Lbl	85	Brown-forman Corporation	750	12	1	[null]	80	1969-11-01 00:00:00	82184090466	82184081693	\$14.53		21.80
6	26827	TENNESSEE WHISKIES	Jack Daniels Old #7 Black Lbl	85	Brown-forman Corporation	1000	12	1	[null]	80	1984-05-01 00:00:00	82184090442	82184081655	\$17.90		26.85

Total rows: 6 Query complete 00:00:00.743

LF Ln 16, Col 1

Note: Products with the same category_name and item_description can have different values for item_no due to differences in packaging attributes (bottle_size, pack, inner_pack), pricing (bottle_price, shelf_price and case_cost), or regulatory listing (list_date).

5. What are the top 10 categories of liquor sold based on the total amount of sales revenue?

Answer: The Top 10 liquor categories by total sales revenue are:

1. Canadian Whiskies – \$16,081,969.16
2. 80 Proof Vodka – \$16,051,839.20
3. Spiced Rum – \$10,539,543.96
4. Imported Vodka – \$7,970,392.87
5. Tequila – \$7,151,416.67
6. Straight Bourbon Whiskies – \$6,999,659.34
7. Whiskey Liqueur – \$6,450,570.64
8. Tennessee Whiskies – \$5,883,598.12
9. Puerto Rico & Virgin Islands Rum – \$4,280,115.93
10. Blended Whiskies – \$4,011,426.50

```
WITH revenue_per_bottle AS (
    SELECT item,
           category_name,
           description,
           total AS total_sales,
           state_btl_cost * bottle_qty AS total_cost,
           CAST(total AS money) - (state_btl_cost * bottle_qty) AS
sales_revenue
    FROM sales
)

SELECT category_name,
       SUM(sales_revenue) AS revenue_per_category
FROM revenue_per_bottle
GROUP BY 1
ORDER BY revenue_per_category DESC
LIMIT 10;
```

	category_name text	revenue_per_category money
1	CANADIAN WHISKIES	\$16,081,969.16
2	80 PROOF VODKA	\$16,051,839.20
3	SPICED RUM	\$10,539,543.96
4	IMPORTED VODKA	\$7,970,392.87
5	TEQUILA	\$7,151,416.67
6	STRAIGHT BOURBON WHISKIES	\$6,999,659.34
7	WHISKEY LIQUEUR	\$6,450,570.64
8	TENNESSEE WHISKIES	\$5,883,598.12
9	PUERTO RICO & VIRGIN ISLANDS RUM	\$4,280,115.93
10	BLENDED WHISKIES	\$4,011,426.50
Total rows: 10		Query complete 00:00:01.218

Revenue is heavily concentrated in **whiskies and vodka categories**, with Canadian whiskies and 80-proof vodka leading almost equally. This indicates that both high-volume, vodka products and widely consumed whisky categories contribute significantly to the overall revenue.

Multiple whisky categories (Canadian, Straight Bourbon, Whiskey Liqueur, Tennessee, and Blended) appear in the Top 10, suggesting that whisky as a broader segment dominates the market in terms of revenue.

Further grouping of these categories and their sales revenue:

Grouping	Revenue
Whisky	\$39,427,223.76
Vodka	\$24,022,232.07
Rum	\$14,819,659.89
Tequila	\$7,151,416.67
Grand Total	\$85,420,532.39

6. Which rum products have sales greater than \$10,000? How about whiskey or vodka products?

Answer: A total of **704 products** across rum, vodka, and whiskey categories recorded sales greater than \$10,000.

The breakdown is as follows:

- Vodka: 397 products mainly by Hawkeye, Smirnoff, Absolut, Grey Goose
- Rum: 182 products mainly by Captain Morgan, Bacardi, Malibu
- Whiskey: 125 products mainly by Crown Royal, Fireball, Seagram's, Canadian Ltd

Vodka accounts for the largest share of products exceeding \$10,000 in sales, representing more than half of the total. This suggests that vodka has the **broadest product participation and strongest depth of demand** across the market.

The results show that a large number of products across all three categories exceed the \$10,000 sales threshold, indicating that demand is **not concentrated in just a few top products**, but distributed across a broad product base.

```

WITH sales_of_product_types AS (
    SELECT item,

        CASE WHEN description ILIKE '%RUM%' THEN 'RUM'
              WHEN category_name ILIKE '%RUM%' THEN 'RUM'

              WHEN description ILIKE '%WHISKEY%' THEN 'WHISKEY'
              WHEN description ILIKE '%WHISKY%' THEN 'WHISKEY'
              WHEN category_name ILIKE '%WHISKEY%' THEN 'WHISKEY'
              WHEN category_name ILIKE '%WHISKY%' THEN 'WHISKEY'

              WHEN description ILIKE '%VODKA%' THEN 'VODKA'
              WHEN category_name ILIKE '%VODKA%' THEN 'VODKA'
              ELSE 'Others'
        END AS product_type,
        SUM(total) AS product_sales
    FROM sales
    GROUP BY item, product_type
    HAVING SUM(total) > 10000
)

SELECT a.product_type,
       a.item,
       b.category_name,
       b.item_description,
       b.pack,
       b.inner_pack,
       b.proof,
       b.list_date,
       a.product_sales
FROM sales_of_product_types a
     LEFT JOIN products b
         ON a.item = b.item_no
WHERE a.product_type <> 'Others'
ORDER BY product_type, product_sales DESC;

```

a. For Rum products

	product_type text	item integer	category_name text	item_description text	pack integer	inner_pack integer	proof text	list_date timestamp without time zone	product_sales numeric
1	RUM	43337	SPICED RUM	Captain Morgan Spiced Rum	12	1	70	1990-10-01 00:00:00	8523370.55
2	RUM	43338	SPICED RUM	Captain Morgan Spiced Rum	6	1	70	1989-05-01 00:00:00	4341131.70
3	RUM	1799	SPICED RUM	Captain Morgan Spiced Barrel	6	1	70	2013-03-01 00:00:00	3774545.06
4	RUM	43336	SPICED RUM	Captain Morgan Original Spiced	12	1	70	1983-11-01 00:00:00	3497803.08
5	RUM	43028	SPICED RUM	Admiral Nelson Spiced Rum	6	1	70	1999-09-01 00:00:00	2164286.62
6	RUM	43127	PUERTO RICO & VIRGIN ISLANDS RUM	Bacardi Superior Rum	12	1	80	1984-05-01 00:00:00	1912471.22
7	RUM	42717	FLAVORED RUM	Malibu Coconut Rum	12	1	42	1994-05-01 00:00:00	1739399.18
8	RUM	43128	PUERTO RICO & VIRGIN ISLANDS RUM	Bacardi Superior Rum	6	1	80	1977-05-01 00:00:00	1703967.50
9	RUM	69947	IMPORTED SCHNAPPS	Rumple Minze Peppermint Schnapps Liqueur	12	1	100	1993-03-01 00:00:00	1670378.69
10	RUM	73055	CREAM LIQUEURS	Rumchata	12	1	27	2014-02-01 00:00:00	1373643.75
11	RUM	43285	SPICED RUM	Captain Morgan Original Spiced Rum Pet	12	1	70	1996-02-01 00:00:00	1334951.04
12	RUM	45278	PUERTO RICO & VIRGIN ISLANDS RUM	Paramount White Rum	6	1	80	1976-11-01 00:00:00	1198708.56
13	RUM	42716	FLAVORED RUM	Malibu Coconut Rum	12	1	42	1994-05-01 00:00:00	1166279.54
Total rows: 704		Query complete 00:00:09.971							

b. For Vodka products

	product_type text	item integer	category_name text	item_description text	pack integer	inner_pack integer	proof text	list_date timestamp without time zone	product_sales numeric
181	RUM	42204	BARBADOS RUM	Blue Chair Bay Banana	12	1	34	2014-07-01 00:00:00	10654.02
182	RUM	46113	PUERTO RICO & VIRGIN ISLANDS RUM	Clearheart Rum	6	1	80	2008-04-01 00:00:00	10051.35
183	VODKA	36308	80 PROOF VODKA	Hawkeye Vodka	6	1	80	1989-10-01 00:00:00	5380753.20
184	VODKA	34007	IMPORTED VODKA	Absolut Swedish Vodka 80 Prf	12	1	80	1989-10-01 00:00:00	3214878.03
185	VODKA	34422	IMPORTED VODKA	Grey Goose Vodka	6	1	80	2002-04-01 00:00:00	3153668.04
186	VODKA	35918	80 PROOF VODKA	Five O'clock Vodka	6	1	80	1977-05-01 00:00:00	2349168.94
187	VODKA	37338	80 PROOF VODKA	UV Vodka PET	6	1	80	2012-12-01 00:00:00	2308126.71
188	VODKA	34006	IMPORTED VODKA	Absolut Swedish Vodka 80 Prf	12	1	80	1980-05-01 00:00:00	2171065.18
189	VODKA	38008	80 PROOF VODKA	Smirnoff Vodka 80 Prf Pet	6	1	80	2004-04-01 00:00:00	2129087.76
190	VODKA	34433	IMPORTED VODKA	Grey Goose Vodka	12	1	80	2008-06-01 00:00:00	1974172.58
191	VODKA	37348	80 PROOF VODKA	Phillips Vodka	6	1	80	2002-02-01 00:00:00	1779996.00
192	VODKA	37996	80 PROOF VODKA	Smirnoff Vodka 80 Prf	12	1	80	1972-02-01 00:00:00	1737681.94
193	VODKA	34008	IMPORTED VODKA	Absolut Swedish Vodka 80 Prf	6	1	80	1988-05-01 00:00:00	1728016.16
Total rows: 704 Query complete 00:00:09.971									

c. For Whiskey products

	product_type text	item integer	category_name text	item_description text	pack integer	inner_pack integer	proof text	list_date timestamp without time zone	product_sales numeric
578	VODKA	40260	FLAVORED VODKA	Smirnoff Wild Honey	12	1	60	2013-10-01 00:00:00	10263.02
579	VODKA	38002	FLAVORED VODKA	Smirnoff Coconut	12	1	70	2013-11-01 00:00:00	10163.98
580	WHISKEY	11297	CANADIAN WHISKIES	Crown Royal Canadian Whisky	12	1	80	1988-08-01 00:00:00	6231299.41
581	WHISKEY	64867	WHISKEY LIQUEUR	Fireball Cinnamon Whiskey	12	1	80	2001-04-01 00:00:00	4758904.71
582	WHISKEY	64866	WHISKEY LIQUEUR	Fireball Cinnamon Whiskey	12	1	80	2001-04-01 00:00:00	3322146.51
583	WHISKEY	11298	CANADIAN WHISKIES	Crown Royal Canadian Whisky	6	1	80	1977-08-01 00:00:00	2497159.27
584	WHISKEY	64868	WHISKEY LIQUEUR	Fireball Cinnamon Whiskey	6	1	66	2011-07-01 00:00:00	2254552.00
585	WHISKEY	25608	BLENDED WHISKIES	Seagrams 7 Crown Bl Whiskey	6	1	80	1977-05-01 00:00:00	2244434.68
586	WHISKEY	86887	WHISKEY LIQUEUR	Southern Comfort	12	1	70	1988-11-01 00:00:00	1873609.92
587	WHISKEY	12408	CANADIAN WHISKIES	Canadian Ltd Whisky	6	1	80	1977-02-01 00:00:00	1757048.16
588	WHISKEY	64858	WHISKEY LIQUEUR	Fireball Cinnamon Whiskey Mini Dispenser	1	60	66	2011-12-01 00:00:00	1186675.02
589	WHISKEY	86886	WHISKEY LIQUEUR	Southern Comfort	12	1	70	1985-05-01 00:00:00	1168521.58
590	WHISKEY	25607	BLENDED WHISKIES	Seagrams 7 Crown Bl Whiskey	12	1	80	1984-05-01 00:00:00	1037201.37
Total rows: 704 Query complete 00:00:09.971									

/*Additional verification queries*/

--Count number of products per category

```

WITH sales_of_product_types AS (
    SELECT item,

        CASE WHEN description ILIKE '%RUM%' THEN 'RUM'
              WHEN category_name ILIKE '%RUM%' THEN 'RUM'

              WHEN description ILIKE '%WHISKEY%' THEN 'WHISKEY'
              WHEN description ILIKE '%WHISKY%' THEN 'WHISKEY'
              WHEN category_name ILIKE '%WHISKEY%' THEN 'WHISKEY'
              WHEN category_name ILIKE '%WHISKY%' THEN 'WHISKEY'

              WHEN description ILIKE '%VODKA%' THEN 'VODKA'
              WHEN category_name ILIKE '%VODKA%' THEN 'VODKA'
              ELSE 'Others'
        END AS product_type,
        SUM(total) AS product_sales
    FROM sales
    GROUP BY item, product_type

```

```

HAVING SUM(total) > 10000
)

SELECT product_type,
       COUNT(*) as number_of_products
FROM sales_of_product_types
WHERE product_type <> 'Others'
GROUP BY product_type
ORDER BY product_type;

```

	product_type text	number_of_products bigint
1	RUM	182
2	VODKA	397
3	WHISKEY	125

Total rows: 3 Query complete 00:00:11.834

7. Which county sold the most amount of vodka during February 2014?

Answer: Polk County sold the highest amount of vodka in February 2014, with a total volume of **180,829,800 ml** (approximately **180,830 liters**).

The term “amount” is interpreted as **physical volume** rather than revenue. To ensure comparability across different bottle sizes, the analysis uses total volume in liters, calculated from the number of bottles sold (*bottle_qty*) and bottle volume (*liter_size*), which is stored in milliliters (ml) in this dataset.

Polk County’s leading position suggests a higher concentration of vodka consumption or distribution during the period, likely influenced by factors such as **population size**, store density, and overall demand concentration.

```

WITH county_consumption AS (
    SELECT county,
           CASE WHEN description ILIKE '%VODKA%' THEN 'VODKA'
                WHEN category_name ILIKE '%VODKA%' THEN
'VODKA'
                ELSE 'Others'
           END AS product_type,
           SUM(liter_size * bottle_qty) AS total_volume_ml
    FROM sales
    WHERE DATE_TRUNC('month', date) = '2014-02-01'
    GROUP BY county, product_type
),

```

```

county_ranking AS (
    SELECT county,
           product_type,
           total_volume_ml,
           RANK() OVER (PARTITION BY product_type ORDER BY
total_volume_ml DESC) as ranking
    FROM county_consumption
    WHERE product_type = 'VODKA'

```

)

```
SELECT *
FROM county_ranking
WHERE ranking = 1;
```

	county text	product_type text	total_volume_ml bigint	ranking bigint
1	Polk	VODKA	180829800	1
Total rows: 1		Query complete 00:00:00.963		

/*What are the characteristics of the county Polk?*/

```
SELECT *
FROM counties
WHERE county = 'Polk'
```

	county [PK] text	population integer
1	Polk	430640

--It is a county with a population size of 430,640

--Check the ranking based on population

```
SELECT *
FROM counties
ORDER BY population DESC
LIMIT 10
```

	county [PK] text	population integer
1	Polk	430640
2	Linn	211226
3	Scott	165224
4	Black Hawk	131090
5	Johnson	130882
6	Woodbury	102172
7	Dubuque	93653
8	Pottawattamie	93158
9	Story	89542
10	Dallas	66135

--Polk is the county with the highest population.

/*Additional verification queries*/

--Check range of dates in February 2014 using TO_CHAR

```
SELECT DISTINCT TO_CHAR(date, 'YYYY-MM-DD') as Feb_dates
FROM sales
WHERE TO_CHAR(date, 'YYYY-MM-DD') between '2014-02-01' AND '2014-02-28'
ORDER BY Feb_dates
```

	feb_dates text
1	2014-02-03
2	2014-02-04
3	2014-02-05
4	2014-02-06
5	2014-02-10
6	2014-02-11
7	2014-02-12
8	2014-02-13
9	2014-02-17
10	2014-02-18
11	2014-02-19
12	2014-02-20
13	2014-02-24
14	2014-02-25
15	2014-02-26
16	2014-02-27

Total rows: 16 Query complete 00:00:02.507

--Range dates are from 2014-02-03 to 2014-02-27

--Check range of dates in February 2014 using DATE_TRUNC

```
SELECT MIN(date) AS min_date, MAX(date) AS max_date
FROM sales
WHERE DATE_TRUNC('month', date) = '2014-02-01'
```

	min_date timestamp without time zone	max_date timestamp without time zone
1	2014-02-03 00:00:00	2014-02-27 00:00:00

Total rows: 1 Query complete 00:00:00.933

--Min and max dates are aligned with the range dates

8. Which counties were in the top 10 counties for vodka sales in any month in 2014? [Top 10 per month]

Answer: There is a **total of 13 counties** which are in the Top 10 counties for vodka sales for at least a month in 2014.

These 7 counties have consistently appeared **every month in the Top 10** for vodka sales in 2014:

- Polk
- Linn
- Scott
- Johnson
- Black Hawk
- Dubuque
- Pottawattamie

These counties demonstrate **consistent, year-round dominance** in vodka sales, rather than one-off spikes. This suggests stable and sustained demand across the year.

```

WITH county_sales AS (
    SELECT county,
           CASE WHEN description ILIKE '%VODKA%' THEN 'VODKA'
                WHEN category_name ILIKE '%VODKA%' THEN
'VODKA'
                ELSE 'Others'
           END AS product_type,
           date,
           DATE_TRUNC('month', date) as sales_month,
           total
    FROM sales
    WHERE DATE_TRUNC('year', date) = '2014-01-01'
),

county_sales_per_month AS (
    SELECT product_type,
           county,
           TO_CHAR(sales_month, 'YYYY-MM') as sales_month,
           SUM(total) AS sales_per_county
    FROM county_sales
    WHERE product_type = 'VODKA'
    GROUP BY county, product_type, sales_month
    ORDER BY sales_month, sales_per_county DESC
),

county_ranking AS (
    SELECT product_type,
           sales_month,
           county,
           sales_per_county,
           RANK() OVER (PARTITION BY sales_month ORDER BY
sales_per_county DESC) as county_rank
    FROM county_sales_per_month
)

SELECT *
FROM county_ranking
WHERE county_rank BETWEEN 1 AND 10
ORDER BY sales_month;

```

	product_type text	sales_month text	county text	sales_per_county numeric	county_rank bigint
1	VODKA	2014-01	Polk	2189476.56	1
2	VODKA	2014-01	Linn	975547.22	2
3	VODKA	2014-01	Scott	764583.12	3
4	VODKA	2014-01	Johnson	688014.54	4
5	VODKA	2014-01	Black Hawk	611497.16	5
6	VODKA	2014-01	Pottawattamie	425965.76	6
7	VODKA	2014-01	Story	358245.36	7
8	VODKA	2014-01	Dubuque	348632.82	8
9	VODKA	2014-01	Woodbury	289273.18	9
10	VODKA	2014-01	Cerro Gordo	222337.54	10
11	VODKA	2014-02	Polk	2260835.86	1
12	VODKA	2014-02	Linn	737865.34	2
13	VODKA	2014-02	Scott	688957.32	3
14	VODKA	2014-02	Johnson	680700.04	4
15	VODKA	2014-02	Black Hawk	562293.56	5
16	VODKA	2014-02	Pottawattamie	311426.32	6
17	VODKA	2014-02	Dubuque	311169.74	7
18	VODKA	2014-02	Story	277883.72	8
19	VODKA	2014-02	Woodbury	272990.38	9
20	VODKA	2014-02	Cerro Gordo	158520.60	10
21	VODKA	2014-03	Polk	2135508.44	1
22	VODKA	2014-03	Linn	839553.50	2
23	VODKA	2014-03	Scott	728768.80	3
24	VODKA	2014-03	Johnson	549395.14	4
25	VODKA	2014-03	Black Hawk	483780.30	5
26	VODKA	2014-03	Pottawattamie	319046.32	6
27	VODKA	2014-03	Dubuque	259820.36	7
28	VODKA	2014-03	Story	257798.26	8
29	VODKA	2014-03	Woodbury	246770.50	9
Total rows: 120		Query complete 00:00:03.567			

9. **Create a report that shows how many times a county appeared in the “top 10 counties for vodka sales in a month” list over the course of 2014.** *[How many times a county appeared in the Top 10 in 2014?]*

Answer: The number of times each county appeared in the top 10 monthly vodka sales rankings in 2014 is as follows:

- 7 Consistent Top Performers (12 out 12 months)
 - Polk
 - Linn
 - Scott
 - Johnson
 - Black Hawk
 - Dubuque
 - Pottawattamie
- 2 Frequent Performers
 - Woodbury – 11 out of 12 months

- Story – 10 out of 12 months
- 2 Occasional performers
 - Cerro Gordo – 7 out of 12 months
 - Dallas – 6 out of 12 months
- Rare Appearances (1 out of 12 months)
 - Des Moines
 - Dickinson

A consistent group of **seven counties consistently dominated vodka sales throughout the year**, appearing in the top 10 every month. This indicates stable and sustained demand concentrated in a small number of regions.

```
WITH county_sales AS (
  SELECT county,
    CASE WHEN description ILIKE '%VODKA%' THEN 'VODKA'
          WHEN category_name ILIKE '%VODKA%' THEN
            'VODKA'
          ELSE 'Others'
    END AS product_type,
    date,
    DATE_TRUNC('month', date) as sales_month,
    total
  FROM sales
  WHERE DATE_TRUNC('year', date) = '2014-01-01'
),

county_sales_per_month AS (
  SELECT product_type,
    county,
    TO_CHAR(sales_month, 'YYYY-MM') as sales_month,
    SUM(total) AS sales_per_county
  FROM county_sales
  WHERE product_type = 'VODKA'
  GROUP BY county, product_type, sales_month
  ORDER BY sales_month, sales_per_county DESC
),

county_ranking AS (
  SELECT product_type,
    sales_month,
    county,
    sales_per_county,
    RANK() OVER (PARTITION BY sales_month ORDER BY
sales_per_county DESC) as county_rank
  FROM county_sales_per_month
),

top_10_vodka_sales AS (
  SELECT *
  FROM county_ranking
  WHERE county_rank BETWEEN 1 AND 10
  ORDER BY sales_month
)
```

```
SELECT county,
        COUNT(*) as top10_frequency
FROM top_10_vodka_sales
GROUP BY county
ORDER BY top10_frequency DESC, county
```

	county text	top10_frequency bigint
1	Black Hawk	12
2	Dubuque	12
3	Johnson	12
4	Linn	12
5	Polk	12
6	Pottawattamie	12
7	Scott	12
8	Woodbury	11
9	Story	10
10	Cerro Gordo	7
11	Dallas	6
12	Des Moines	1
13	Dickinson	1

Total rows: 13 Query complete 00:00:03.393

10. What is the trend of sales by month? Break up variables such as bottle_price or liter_size into categories (for example: cheap, medium, or expensive).

Answer: Monthly liquor sales show a clear **seasonal trend**, with sales **peaking in the first half** of the year especially in April to May, **declining during mid-year** from June to August, and **partially recovering towards the later months** before **dropping sharply in December**.

The **dip in December across all segments** may warrant further investigation, as it deviates from typical retail seasonality patterns, such as Christmas season and year-end celebrations, and could indicate data coverage gaps or market-specific behaviour.

When segmented by price category—**Value (<\$10)**, **Standard (\$10–\$30)**, and **Premium (>\$30)**—distinct patterns emerge across different price tiers.

a. Overall trend of sales by month

```
SELECT TO_CHAR(date, 'YYYY-MM') as sales_month,
        SUM(total) as sales_per_month
FROM sales
GROUP BY sales_month
ORDER BY sales_month;
```

	sales_month text	sales_per_month numeric
1	2014-01	40331023.96
2	2014-02	38543588.42
3	2014-03	38603543.82
4	2014-04	45388494.64
5	2014-05	42110068.64
6	2014-06	24414597.17
7	2014-07	22096736.93
8	2014-08	21207347.84
9	2014-09	23440959.93
10	2014-10	26752602.20
11	2014-11	20244210.54
12	2014-12	9306222.86
13	2015-01	18558276.11
14	2015-02	21295350.55

Total rows: 14 Query complete 00:00:01.570

Total sales peak from **April to May**, followed by a noticeable decline through the summer months, a mild recovery in September–October, and a significant drop in December.

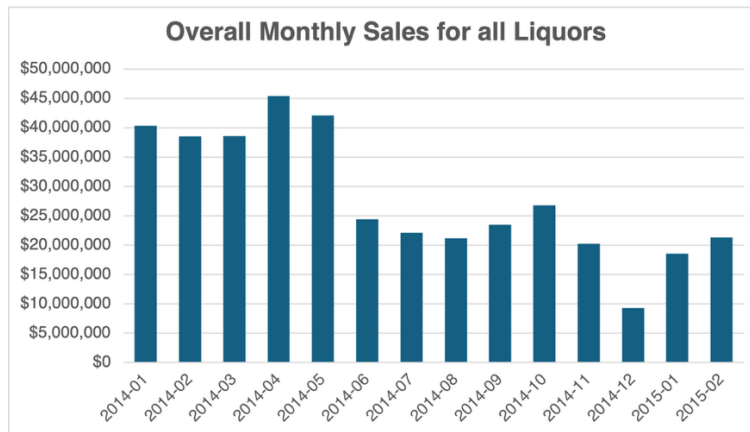


Chart 1: Monthly Sales from January 2014 to February 2015

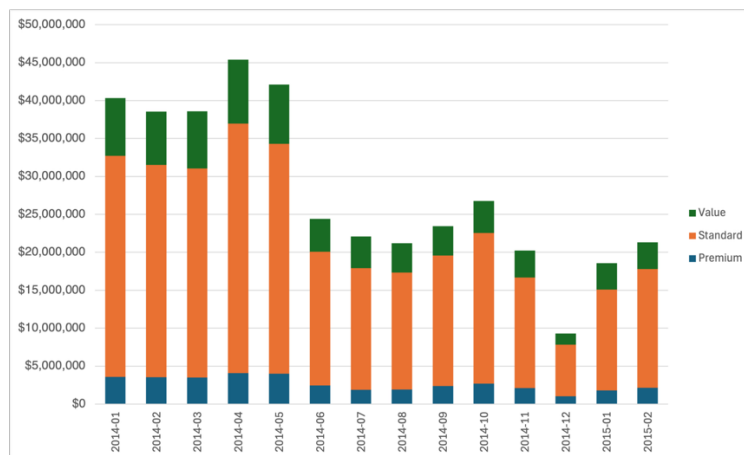


Chart 2: Monthly Sales by Bottle Price Category

b. Trend of sales by month by bottle price category

```
SELECT CASE WHEN btl_price <= CAST(10.0 AS money) THEN 'Value'
           WHEN btl_price <= CAST(30.0 AS money) THEN
'Standard'
           ELSE 'Premium'
        END AS bottle_size_category,
        TO_CHAR(date, 'YYYY-MM') as sales_month,
        SUM(total) as sales_per_month
FROM sales
GROUP BY bottle_size_category, sales_month;
```

	bottle_size_category text	sales_month text	sales_per_month numeric
1	Premium	2014-01	3571601.48
2	Premium	2014-02	3532139.06
3	Premium	2014-03	3477212.86
4	Premium	2014-04	4081471.84
5	Premium	2014-05	4000804.62
6	Premium	2014-06	2458592.22
7	Premium	2014-07	1878994.74
8	Premium	2014-08	1907568.95
9	Premium	2014-09	2359686.10
10	Premium	2014-10	2687414.01
11	Premium	2014-11	2114187.76
12	Premium	2014-12	1023949.25
13	Premium	2015-01	1790421.28
14	Premium	2015-02	2155628.91
15	Standard	2014-01	29131623.64
16	Standard	2014-02	27971726.26
17	Standard	2014-03	27589251.70
18	Standard	2014-04	32895961.04
19	Standard	2014-05	30299662.40
20	Standard	2014-06	17629792.62
21	Standard	2014-07	16031618.90
22	Standard	2014-08	15409688.50
23	Standard	2014-09	17218335.93
24	Standard	2014-10	19876768.67
25	Standard	2014-11	14548541.68
26	Standard	2014-12	6785629.52
27	Standard	2015-01	13294111.68
28	Standard	2015-02	15642633.25
Total rows: 42		Query complete 00:00:02.756	

The **Value segment**, with bottle prices less than \$10 shows relatively stable sales patterns and closely mirrors the overall trend, though at a lower scale. These provide **consistent baseline demand** and are **less volatile than Premium products**.



Chart 3: Monthly Sales of Value Liquors

The **Standard segment**, with bottle prices from \$10 to \$30 consistently contributes the largest share of monthly sales across all periods. This suggests that this is the **core market segment** balancing between affordability and brand quality.

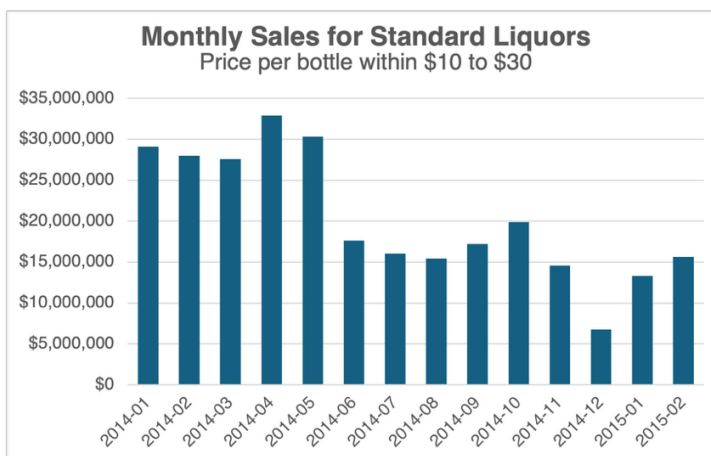


Chart 4: Monthly Sales of Standard Liquors

The **Premium segment, with bottle prices more than \$30** contributes the smallest share of sales but maintains a consistent pattern aligned with overall trends. Premium products have **lower volume but steady demand**, reinforcing their role as a niche but stable segment.

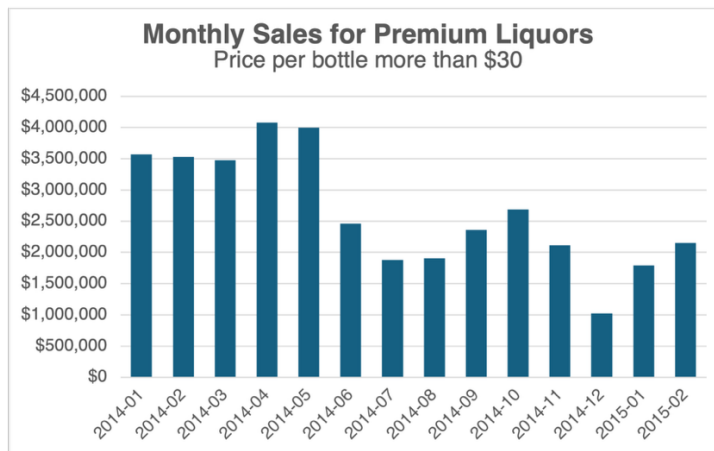


Chart 5: Monthly Sales of Premium Liquors

11. Pick one of these 4 questions:

a. How many stores have more than \$2,000,000 in total sales?

Answer: 24 out of 1,973 stores (approximately 1.2%) have total sales exceeding \$2,000,000.

A relatively small proportion of stores account for very high sales volumes, indicating that **revenue is highly concentrated among top-performing outlets**.

Of these, **16 out of 24 stores (67%)** belong to large-format retail chains such as **Hy-Vee, Sam's Club, and Costco**, which typically benefit from higher customer traffic and bulk purchasing behaviour.

There is also a strong geographic concentration. **17 out of the 24 stores (71%)** are located in the top-performing counties for vodka sales—**Polk, Linn, Johnson, Scott, Black Hawk, Dubuque, and Pottawattamie** (refer to Question 9).

Overall, this suggests that both **store format (large-scale retailers)** and **location (high-demand regions)** are key drivers of total sales performance.

--Count of stores

```
SELECT COUNT(*) as number_of_stores
FROM (
    SELECT store
    FROM sales
    GROUP BY store
    HAVING SUM(total) > 2000000
) stores_with_sales_2M
```

--24 stores

	number_of_stores bigint
1	24
Total rows: 1 Query complete 00:00:00.932	

--What are those stores?

```

SELECT a.store,
       b.name,
       b.store_status,
       regexp_replace(b.store_address, '[\r\n]+', ' ', 'g') as
store_address, --Replace all line breaks in store_address with a space
       a.county,
       SUM(total) AS store_sales
FROM sales a
JOIN stores b
  USING (store)
GROUP BY a.store, b.name, b.store_status, b.store_address, a.county
HAVING SUM(total) > 2000000
ORDER BY store_sales DESC;

```

	store integer	name text	store_status text	store_address text	county text	store_sales numeric
1	2633	Hy-vee #3 / Bdi / Des Moines	A	3221 Se 14th St Des Moines, IA 503200000 (41.55407025200003, -93.59676391599999)	Polk	13920087.22
2	4829	Central City 2	A	1501 Michigan Ave Des Moines, IA 50314 (41.60583075100004, -93.61375431199997)	Polk	11942399.97
3	3420	Sam's Club 6344 / Windsor Heights	A	1101 73rd Street Windsor Heights, IA 503110000 (41.597748839000076, -93.71836318499999)	Polk	6159480.06
4	3385	Sam's Club 8162 / Cedar Rapids	A	2605 Blairs Ferry Rd Ne Cedar Rapids, IA 524020000 (42.03468354000006, -91.68021044599999)	Linn	5734721.57
5	2512	Hy-vee Wine and Spirits / Iowa City	A	1720 Waterfront Dr Iowa City, IA 522400000 (41.64292615900007, -91.53064887599999)	Johnson	5665143.70
6	3814	Costco Wholesale #788	A	7205 Mills Civic Pkwy West Des Moines, IA 502660000 (41.56158773800007, -93.80428295799999)	Dallas	4907465.88
7	3952	Lot-a-spirits	A	3780 State St Bettendorf, IA 527220000 (41.529565827000056, -90.48082335099997)	Scott	4289169.59
8	3354	Sam's Club 8238 / Davenport	A	3845 Elmore Ave. Davenport, IA 528070000 (41.55976901200006, -90.52713019399994)	Scott	3308625.56
9	2625	Hy-Vee Wine and Spirits #2	A	3301 W Kimberly Rd Davenport, IA 528040000 (41.56012403400007, -90.62359906599994)	Scott	3169984.14
10	3773	Benz Distributing	A	501 7th Ave Se Cedar Rapids, IA 524010000 (41.975795025000025, -91.65975629199994)	Linn	3129506.57
11	2599	Hy-vee Wine and Spirits / Coralville	A	Lantern Park Plaza Coralville, IA 522410000 (41.677040815000055, -91.58666630699997)	Johnson	2920927.81
12	3494	Sam's Club 6514 / Waterloo	A	210 East Tower Park Dr Waterloo, IA 507020000 (42.45634219000004, -92.35346388299996)	Black Hawk	2862977.89
13	3447	Sam's Club 6432 / Sioux City	A	4201 S. York St. Sioux City, IA 511060000 (42.433921534000035, -96.37165902399994)	Woodbury	2652494.66
14	3692	Wilkie Liquors	A	724 1st St E Mt Vernon, IA 523140000 (41.91830538100004, -91.41044168499997)	Linn	2534347.62
15	3524	Sam's Club 6568 / Ames	A	305 Airport Rd Ames, IA 500100000 (42.00109798200003, -93.61356072799998)	Story	2523731.14
16	4677	Costco Wholesale #1111 / Coralville	A	2900 Heartland Dr Coralville, IA 52241 (41.69794646400004, -91.61526092999998)	Johnson	2402474.38
17	2616	Hy-vee Food and Drug / Clinton	A	901 South 4th St Clinton, IA 527320000 (41.83645241900007, -90.19318277899998)	Clinton	2395404.67
18	3820	Charlie's Wine and Spirits,	A	507 W 19th St Sioux City, IA 511030000 (42.51056671100008, -96.42021156999999)	Woodbury	2299523.69
19	2629	Hy-vee Food Store #2 / Council Bluff	A	1745 Madison Ave Council Bluffs, IA 515030000 (41.24353036400004, -95.82661436399997)	Pottawattamie	2256248.69
20	4312	I-80 Liquor / Council Bluffs	A	2411 S 24th St #1 Council Bluffs, IA 515010000 (41.23810075100005, -95.87915418699998)	Pottawattamie	2163609.13
21	2593	Hy-vee Food Store / Carroll	A	905 Us Highway 30 West Carroll, IA 514010000 (42.07083983900003, -94.87687526699995)	Carroll	2154993.97
22	3926	Liquor Downtown / Iowa City	A	315 S Gilbert St Iowa City, IA 522400000 (41.657440301000065, -91.53031701899994)	Johnson	2138974.69
23	2648	Hy-vee #4 / Wdm	A	555 S 51st St West Des Moines, IA 502650000 (41.56037553800007, -93.77292706499998)	Polk	2072855.15
24	2506	Hy-vee #1044 / Burlington	A	3140 Agency Burlington, IA 526010000 (40.81466829500005, -91.13590017199994)	Des Moines	2055957.16
Total rows: 24		Query complete 00:00:03.820				

BONUS

2. **Store 2238 (Adventureland Inn at 3200 Adventureland Dr) sold \$883.24 in April and \$27,526.38 in May, for a 3017% growth rate. That was the highest percentage month-on-month growth rate. Create a query that shows this and the next 9 highest after that.**

Answer: The store with the highest month-on-month growth rate is **Adventureland Inn (Store 2238)**

- April 2014 Sales: **\$883.24**
- May 2014 Sales: **\$27,526.38**
- **Growth Rate: 3016.52%**

This is the highest percentage increase observed. The query also identifies the next nine highest growth instances across all stores.

Such extreme growth rates are typically driven by **very low base values in the previous month**, where even moderate increases in absolute sales result in disproportionately large percentage changes.

As a result, high percentage growth does not necessarily indicate sustained performance improvement. Instead, it may reflect **short-term spikes, promotional activity, or recovery from unusually low prior sales**. Therefore, growth rates should be interpreted alongside absolute sales values to avoid misleading conclusions.

```
WITH monthly_store_sales AS (  
    SELECT  
        TO_CHAR(date, 'YYYY-MM') as sales_month,  
        store,  
        SUM(total) AS monthly_sales  
    FROM sales  
    GROUP BY store, sales_month  
    ORDER BY sales_month, store  
) ,  
  
MoM_sales_growth AS (  
    SELECT store,  
        sales_month,  
        monthly_sales,  
        LAG(monthly_sales) OVER (PARTITION BY store ORDER BY  
sales_month) AS previous_month_sales  
    FROM monthly_store_sales  
) ,  
  
sales_growth_rate AS (  
    SELECT store,  
        sales_month,  
        previous_month_sales,  
        monthly_sales,  
        ROUND(((monthly_sales - previous_month_sales) /  
previous_month_sales ) * 100,2) AS growth_rate_percentage  
    FROM MoM_sales_growth  
    WHERE previous_month_sales IS NOT NULL --exclude the first month  
of sales | no comparison
```

```

        AND previous_month_sales > 0 --exclude months with no sales |
divide by 0 error
)

```

```

SELECT a.store,
       b.name,
       b.store_address,
       a.sales_month,
       a.previous_month_sales,
       a.monthly_sales,
       a.growth_rate_percentage
FROM sales_growth_rate a
JOIN stores b
     USING (store)
ORDER BY growth_rate_percentage DESC
LIMIT 10;

```

	store integer	name text	store_address text	sales_month text	previous_month_sales numeric	monthly_sales numeric	growth_rate_percentage numeric
1	2238	Adventureland Inn	3200 Adventureland Dr	2014-05	883.24	27526.38	3016.52
2	5050	Fareway Stores #147 / Carlisle	1115, Bluestem Dr	2015-01	1696.54	26055.32	1435.79
3	4275	Fareway Stores #980 / Knoxville	1308 S Lincoln St	2014-09	2925.30	41674.59	1324.63
4	4169	Super Quick 2 / Hubbell	1824 Hubbell Ave	2014-11	985.92	12794.70	1197.74
5	4255	FAREWAY STORES #058 / ORANGE CITY	512 8th Se	2015-01	1598.77	20663.34	1192.45
6	4669	Vom Fass / Des Moines	833, 42nd St	2014-10	649.24	7231.01	1013.77
7	3866	Target Store T-1170 / Ames	320 S Duff	2014-03	3301.92	35800.68	984.24
8	4717	Walgreens #05512 / Bettendorf	3425 Middle Rd	2014-08	1412.70	14064.95	895.61
9	4104	Fareway Stores #491 / Mason City	400 North Delaware Ave	2015-01	2150.64	20584.26	857.12
10	4248	Fareway Stores #021 / Sheldon	2603 Park St	2015-01	837.18	7799.22	831.61
Total rows: 10		Query complete 00:00:01.828					LF

REFERENCES:

Data Dictionaries:

https://data.iowa.gov/Sales-Distribution/Iowa-Liquor-Products/gckp-fe7r/about_data

https://data.iowa.gov/Regulation/Iowa-Liquor-Stores/ykb6-ywnd/about_data

https://data.iowa.gov/Community-Demographics/County-Population-in-Iowa-by-Year/qtnr-zsrc/about_data

https://data.iowa.gov/Sales-Distribution/Iowa-Liquor-Sales/m3tr-ghgy/about_data